

LOGIC No: SEQ-7801	DESCRIPTION: COOLING TOWER FAN (CT-7801) SHUTDOWN LOGIC	SIL: SIL 1	TAG: CT-7801 FLOC: TJC-LLD-7800-01
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CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP FAN MOTOR CT-7801	EFF-2 ANNUNCIATE ALARM ON DCS	EFF-3 REQUEST START OF SPARE CELL CT-7802
T1	Fan/gearbox vibration HIGH-HIGH	VSHH-7802	> 9 mm/s	1oo1	X	X	X
T2	Gearbox oil temperature HIGH-HIGH	TSHH-7803	> 90 degC	1oo1	X	X	X
T3	Gearbox oil pressure LOW	PSL-7807	< 0.8 barg	1oo1	X	X	
T4	Fan motor overload	MPR-7801	MPR pickup	1oo1	X	X	X

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP FAN MOTOR CT-7801
EFF-2	ANNUNCIATE ALARM ON DCS
EFF-3	REQUEST START OF SPARE CELL CT-7802

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Gearbox oil level OK	LG gearbox
2	Basin level normal (> LSL-7804)	LT-7804
3	No fan lockout / maintenance key released	lockout
=>	ALL permissives TRUE (AND) => START CT-7801 ENABLED	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-7801 for instrument loops and the Datasheet for design limits of CT-7801.